**StereoLearn: An Online Platform for Learning and Automatic Grading of Organic Structure Representations**

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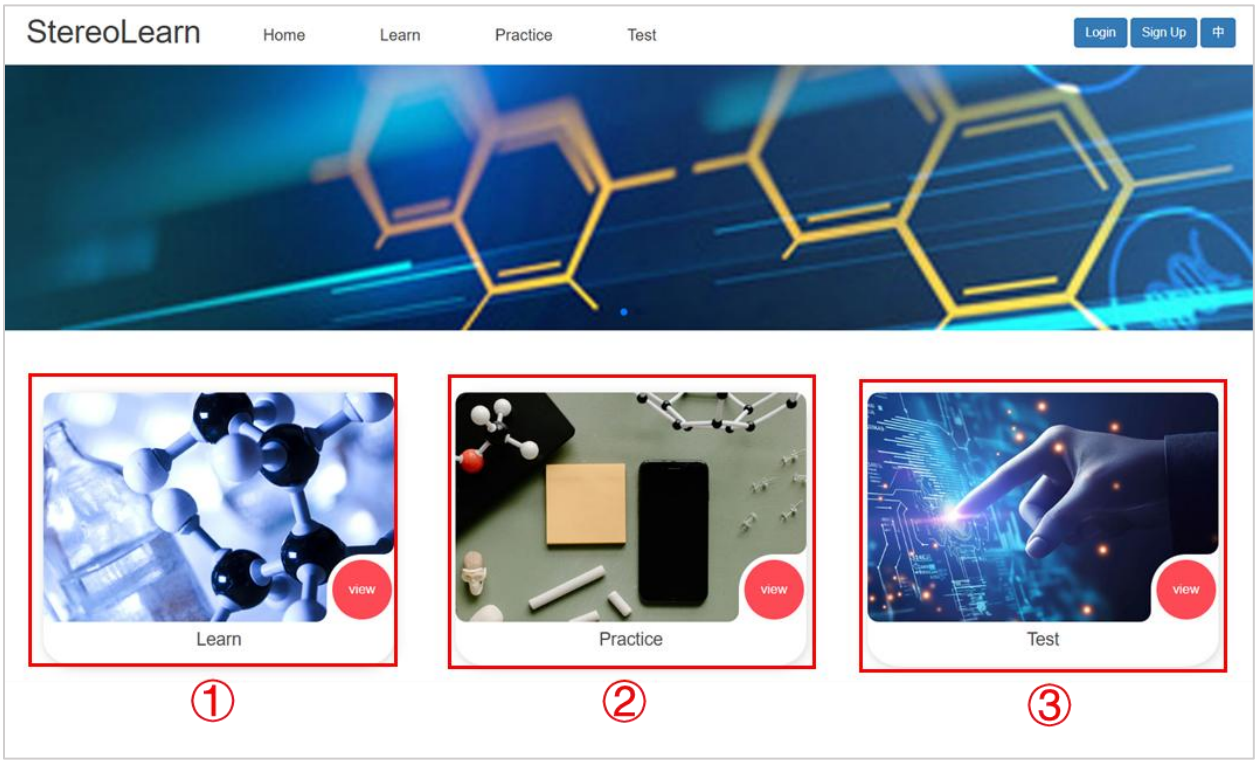
# User Guide

1. **Website URL**: https://en.stereolearn.chemcognitive.online/




The StereoLearn landing page features the three core modules of the platform: *Learn*, *Practice* and *Test*.

## 2. Account Types and Access

StereoLearn supports three levels of access: Regular, Instructor, and Administrator.

- Regular Accounts: Allow users to engage with learning content and complete assignments in the *Practice* or *Test* modules.
- Instructor/Administrator Accounts: Required for releasing homework, managing the backend database, and configuring system settings.

Guest Access: Guests can explore the platform's functions using the following credentials:

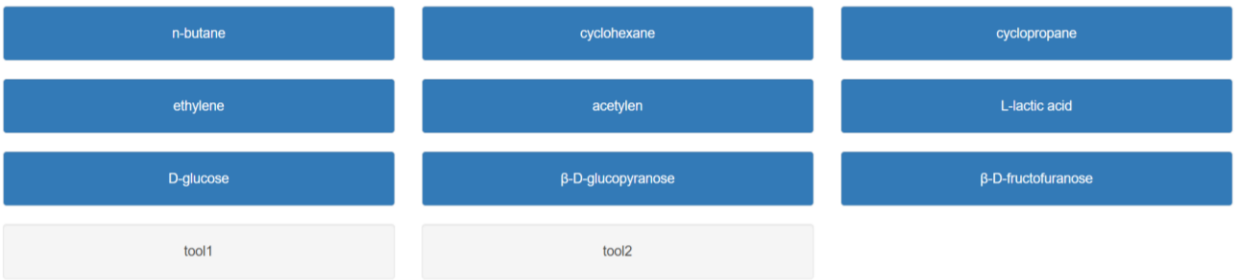
- Username: guest
- Password: guest01

Registration: Individual users may register using their email address. For batch registration or to request Instructor/Administrator privileges, please contact the author at xiaoli@pku.edu.cn.

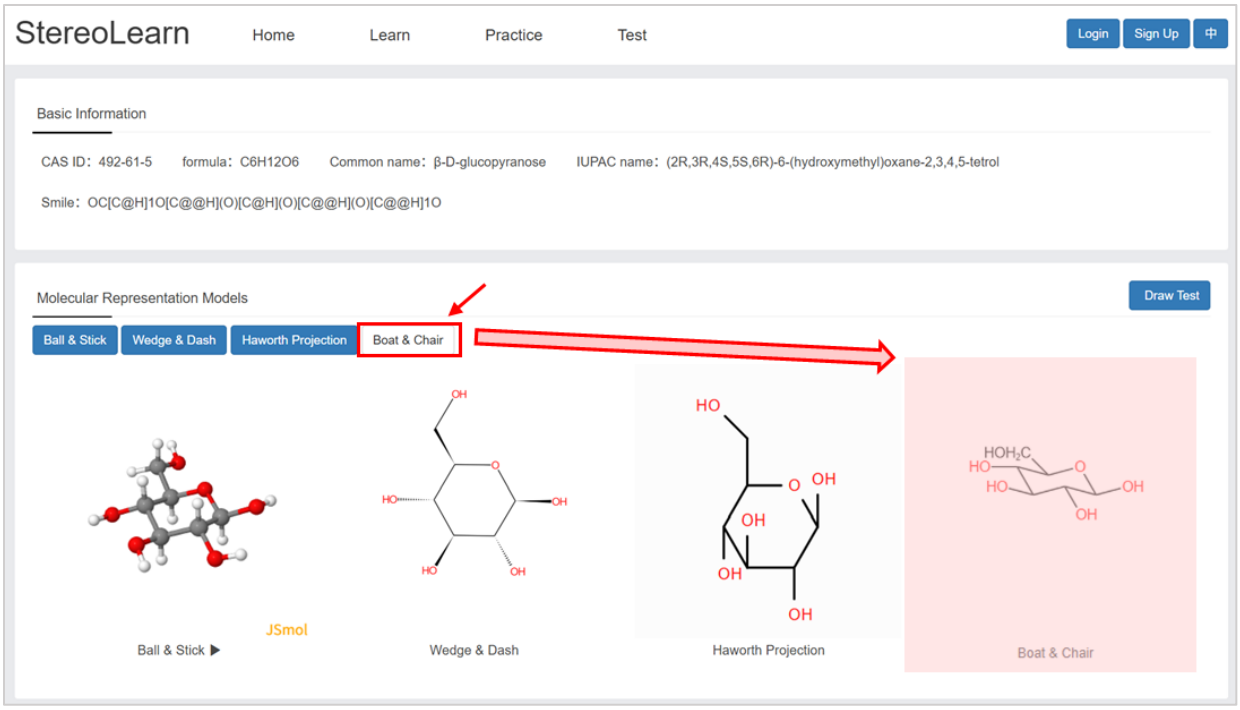
## 3. Core Modules

### ① *Learn* Module

The *Learn* module is publicly accessible and does not require a login. This page provides links to representative molecular structures and essential external tools for chemical drawing. Administrators have the authority to manage the molecular identities and associated data displayed in this module.




Example: β-D-glucopyranose

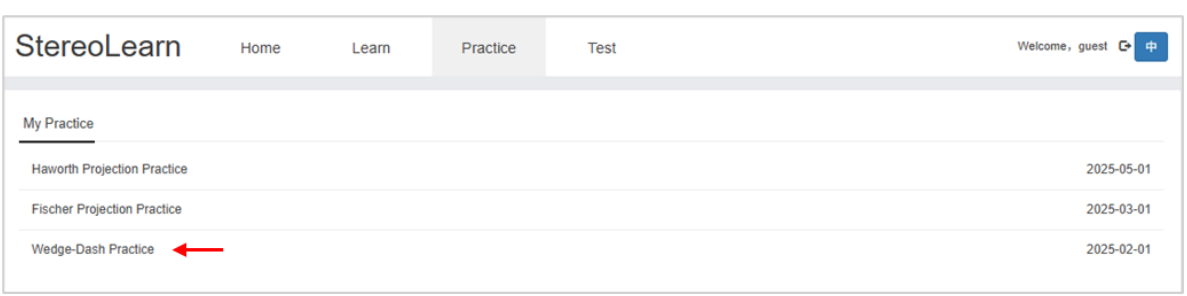



When loading a specific molecule, the system displays a basic information card, an interactive 3D ball-and-stick model, and the standard dash-wedge structure by default. Users can view alternative diagrammatic conventions—such as Haworth, Fischer, or Newman projections, and chair-and-boat conformations—by clicking the corresponding icons.

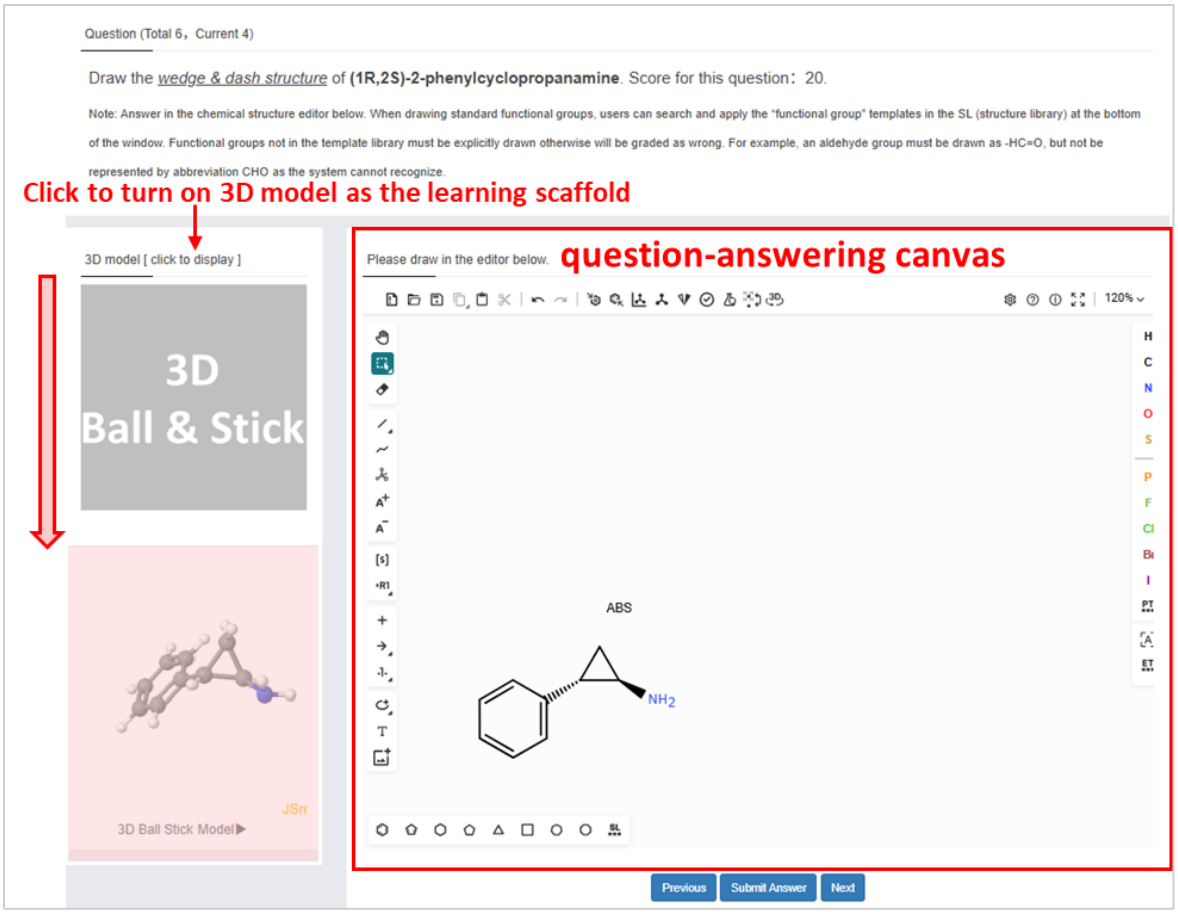
*Note: To aid effective spatial visualization, only pedagogically appropriate drawing styles are available for each molecule. For instance, β-D-glucopyranose does not include Fischer or Newman projections, as these are not standard representations for this structure.*

### ② *Practice* Module

The *Practice* module hosts online structure-drawing exercises and an interactive answering platform. Users must be logged in to access these functions. Upon logging in, a list of available practice sets will be displayed.



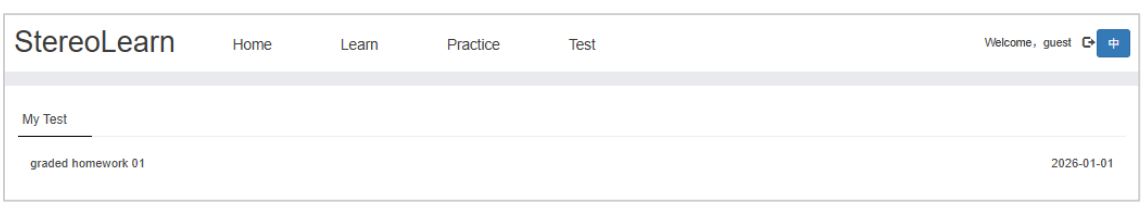

Below shows the screenshot of the “Wedge-Dash Practice” question 4 as an example.



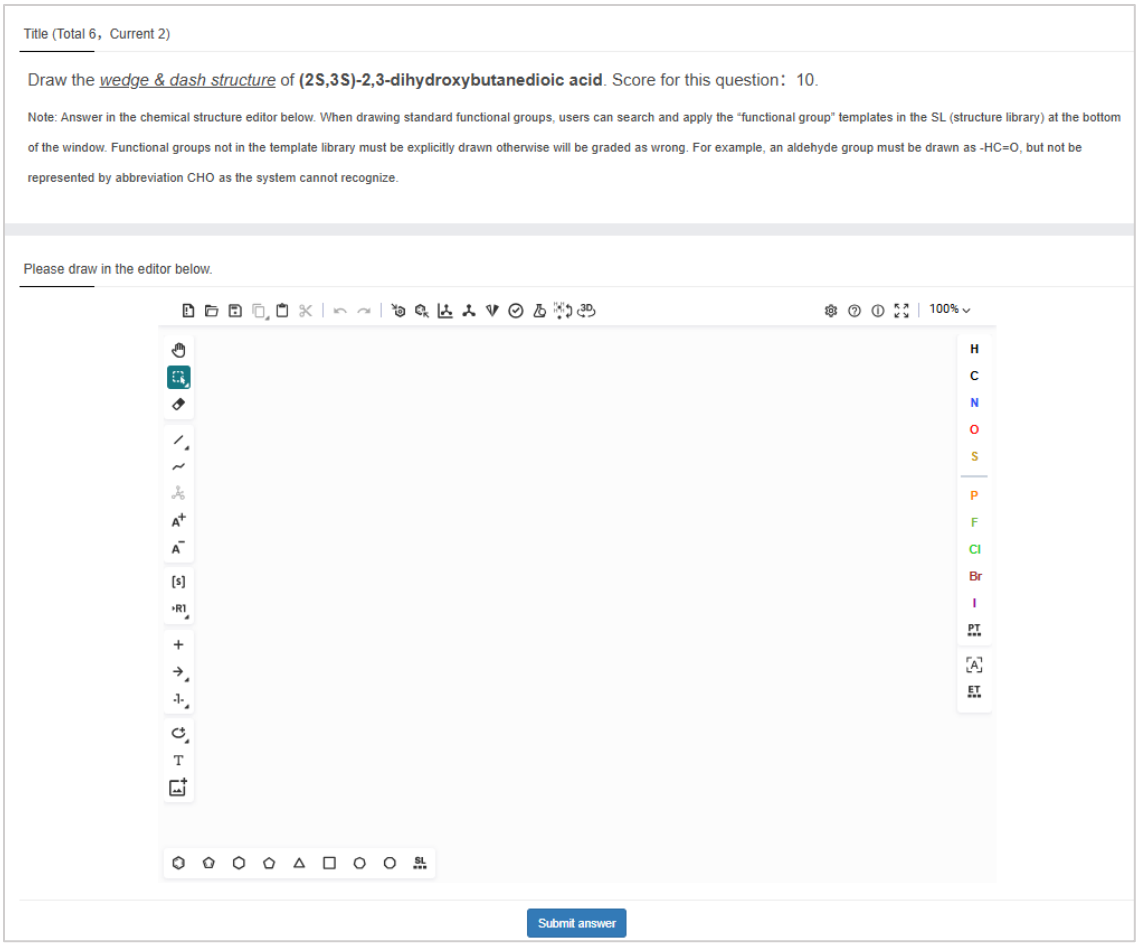

The primary workspace consists of a question prompt at the top and a drawing canvas in the lower-right section. Unique to the *Practice* module is a 3D ball-and-stick model, accessible via a toggle on the left side of the canvas to assist in spatial reasoning. Users can submit and grade their answers multiple times to gain formative feedback. The module allows for free navigation (forward and backward) within a practice set to encourage mastery through repetition.

### ③ *Test* Module

The *Test* module is designed for formal assessment. Similar to the *Practice* module, a login is required to view assigned homework or examinations.




Below shows the screenshot of question 2 in the "graded homework 01" as an example.




To evaluate a student's independent representational competency, the interactive 3D visualization scaffolding found in the *Practice* module is disabled in this environment. Furthermore, the navigation and submission protocols are more restrictive to maintain academic integrity. Each question allows for only a single submission. Users must proceed through the test set in a linear fashion; once a question is submitted, users cannot return to previous items.